Discrete diffusion as posterior learning

1 Discrete diffusion as posterior learning

Fix a finite state space S. This handout takes the diffusion unit's bridge-first viewpoint and specializes it to a setting where the reverse-time object can sometimes be represented exactly. The central ingredients are the endpoint laws p_0 and p_1, the chosen bridge $K_{t|0}(\cdot \mid x_0)$, and the endpoint posterior $q_{0|t}$ induced by that bridge. In finite state spaces the resulting question becomes unusually clean: when does reverse sampling reduce to classification, and when does tractability require a change in the bridge itself?

Let p_0 be the data law on S, let p_1 be a reference law, and let $K_{t|0}(\cdot \mid x_0)$ be a conditional probability path with

$$K_{0|0}(\cdot \mid x_0) = \delta_{x_0}, \qquad K_{1|0}(\cdot \mid x_0) = p_1.$$

Averaging over $X_0 \sim p_0$ gives the marginal law p_t, and Bayes' rule gives the endpoint posterior

$$q_{0|t}(x_0 \mid x) = \frac{p_0(x_0) K_{t|0}(x \mid x_0)}{p_t(x)}.$$

If we knew this posterior exactly, reverse sampling would be straightforward: sample an endpoint from $q_{0|t}(\cdot \mid x)$ and then take the bridge step conditioned on that endpoint. This is the finite-state specialization of the posterior-averaging mechanism developed in the diffusion-models unit.

When S is small, the one-hot conditional expectation $\mathbb{E}[e_{X_0} \mid X_t = x]$ is already the full posterior vector $q_{0|t}(\cdot \mid x)$. For a one-token vocabulary model that means classifier outputs can represent the exact reverse-time object. For the full sequence state space $S = \mathcal{V}^d$, the same statement remains true but becomes computationally useless because the simplex has size $|\mathcal{V}|^d$. Practical discrete diffusion therefore changes the bridge, not the posterior-averaging principle: it replaces the exact full-sequence bridge by a coordinatewise bridge whose reverse generator depends only on coordinatewise posterior marginals. The discussion below follows that progression from the exact finite-state bridge to the exact full-sequence model and then to the factorized sequence bridge used in practice.

1.1 Exact posterior learning on a finite state space

We now specialize the bridge and posterior notation of the diffusion-models unit to a finite state space S. Let p_0 be the data law, p_1 a reference law, and

$$K_{t|0}(\cdot \mid x_0), \qquad K_{0|0}(\cdot \mid x_0) = \delta_{x_0}, \qquad K_{1|0}(\cdot \mid x_0) = p_1$$

is a conditional probability path. The corresponding marginal path is

$$p_t(x) = \sum_{x_0 \in S} p_0(x_0)\, K_{t|0}(x \mid x_0),$$

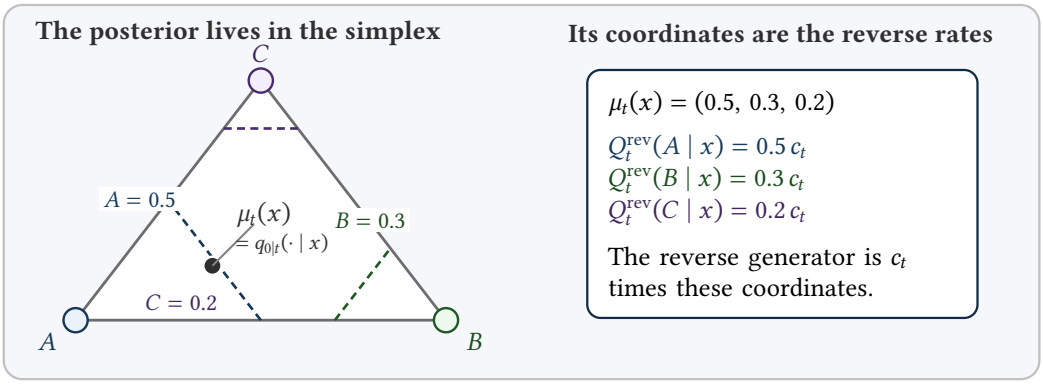

Figure 1: In the one-token setting, the posterior is literally a point in the simplex over endpoint states. Learning that point is genuinely distributional, and the reverse generator just rescales its coordinates by the common jump rate c_t.

and Bayes' rule gives the posterior over the $t = 0$ endpoint states,

$$q_{0|t}(x_0 \mid x) = \frac{p_0(x_0)K_{t|0}(x \mid x_0)}{p_t(x)}.$$

This formula is understood on the support of p_t, that is, for states x with $p_t(x) > 0$.

In the Euclidean development of the diffusion-models unit, one passes from the full posterior to conditional expectations that are infinitesimally sufficient. On a finite state space, a one-hot embedding of the $t = 0$ endpoint loses no information, so the same conditional expectation is the full posterior itself. Suppressing the dependence on S, write $e_z \in \mathbb{R}^{|S|}$ for the basis vector indexed by $z \in S$.

Proposition 1.1 (The one-hot conditional expectation is the full posterior)**.** *Define*

$$\mu_t(x) := \mathbb{E}[e_{X_0} \mid X_t = x] \in \mathbb{R}^{|S|}.$$

Then

$$\mu_t(x)_z = q_{0|t}(X_0 = z \mid X_t = x), \qquad z \in S. \tag{1}$$

Proof. Because $e_{X_0} = e_z$ on the event $\{X_0 = z\}$,

$$\mu_t(x) = \sum_{z \in S} e_z\, \mathbb{P}(X_0 = z \mid X_t = x).$$

Taking the zth coordinate gives (1). ■

Proposition 1.1 is the finite-state version of the Euclidean expectation reduction, except that here the reduction is exact. In Euclidean space, quantities like $\mathbb{E}[\dot{X}_t \mid X_t = x]$ or $\mathbb{E}[X_0 \mid X_t = x]$ summarize a richer posterior. For a finite state space, the one-hot expectation $\mathbb{E}[e_{X_0} \mid X_t = x]$ already is the posterior. When $S = \mathcal{V}$, a classifier over the vocabulary is therefore an exact posterior model.

This is the discrete analogue of the conditional-versus-marginal story in the diffusion-models unit. We observe the current bridge state $X_t = x$, infer the conditional law of the endpoint $X_0 \mid X_t = x$, and then use

a sampled endpoint to take the reverse bridge step. The only difference from the Euclidean case is that on a finite set the relevant posterior can sometimes be represented exactly by a finite vector.

Operationally, an exact finite-state bridge can therefore be sampled in two stages. Given a current state x_t and an earlier time $s < t$, first draw an endpoint and then draw from the bridge conditioned on that endpoint:

$$\tilde{X}_0 \sim q_{0|t}(\cdot \mid x_t), \qquad X_s \sim K_{s|t}^{\tilde{X}_0}(\cdot \mid x_t).$$

Marginalizing $\tilde{X}_0$ recovers the exact earlier-time kernel $K_{s|t}(\cdot \mid x_t)$. So in the one-token case there is no approximation to explain away: infer the full conditional law of $X_0 \mid X_t = x_t$, sample an endpoint from it, and then take the exact bridge step conditioned on that sample.

Proposition 1.2 (Endpoint-conditioned bridge marginalization on a finite state space)**.** *Let $0 \le s \le t \le 1$. Write $K_{s|t}^{x_0}(\cdot \mid x_t)$ for the bridge kernel from time t to the earlier time s conditioned on the endpoint $X_0 = x_0$. Then*

$$K_{s|t}(y \mid x_t) = \sum_{x_0 \in S} K_{s|t}^{x_0}(y \mid x_t)\, q_{0|t}(x_0 \mid x_t).$$

Proof. For states x_t with $p_t(x_t) > 0$, the law of total probability gives

$$\mathbb{P}(X_s = y \mid X_t = x_t) = \sum_{x_0 \in S} \mathbb{P}(X_s = y \mid X_0 = x_0, X_t = x_t)\, \mathbb{P}(X_0 = x_0 \mid X_t = x_t).$$

This is exactly the displayed identity. ■

Proposition 1.3 (Exact reverse kernels transport the marginal path)**.** *For every $0 \le s \le t \le 1$,*

$$p_s(y) = \sum_{x \in S} p_t(x)\, K_{s|t}(y \mid x), \qquad y \in S.$$

Proof. Again by the law of total probability,

$$p_s(y) = \sum_{x \in S} \mathbb{P}(X_t = x)\, \mathbb{P}(X_s = y \mid X_t = x) = \sum_{x \in S} p_t(x)\, K_{s|t}(y \mid x).$$

■

The diffusion-models unit combines these two identities into a domain-agnostic exactness theorem: if at every grid time t_k we can sample $\tilde{X}_0 \sim \mathrm{Law}(X_0 \mid X_{t_k} = x)$ from the exact endpoint posterior and then sample the reverse step from $K_{t_{k-1}|t_k}^{\tilde{X}_0}(\cdot \mid x)$, then reverse sampling is exact on any grid. On a finite state space the same statement applies verbatim. So for the exact one-token or full-sequence model there is no approximation to explain away: once the full posterior is exact, every reverse step uses the exact bridge kernel. The generator formulas below are the infinitesimal version of that general result.

To make that infinitesimal picture concrete, consider the finite-state mixture path

$$K_{t|0}(x \mid x_0) = \alpha_t\, \delta_{x_0}(x) + (1 - \alpha_t)\, r(x), \qquad x, x_0 \in S, \tag{2}$$

where r is a reference law on S, and $\alpha : [0,1] \to [0,1]$ is differentiable and nonincreasing with $\alpha_0 = 1$ and $\alpha_1 = 0$. This is the simplest categorical bridge on a one-token state space. If we instead take $S = \mathcal{V}^d$, then

(2) is a mathematically exact non-factorized sequence model: the state space is all endpoint sequences, and the learned output would be a posterior over complete sequences rather than over individual coordinates.

A useful coupling introduces a single switch time. Draw τ with survival function $\mathbb{P}(\tau > t) = \alpha_t$, draw $\Xi \sim r$ independently, and set

$$X_t = \begin{cases} X_0, & t < \tau, \\ \Xi, & t \geq \tau. \end{cases} \tag{3}$$

Then (3) realizes exactly the bridge (2). To describe reverse-time motion infinitesimally, we use a time-dependent rate matrix Q_t, meaning that over a short backward step of length h,

$$K_{t-h|t}(y \mid x) = \delta_x(y) + hQ_t(y \mid x) + o(h).$$

So the off-diagonal entries $Q_t(y \mid x)$ are first-order jump intensities from x to y.

Proposition 1.4 (Full-state generator rates are posterior probabilities)**.** *Let $K_{t|0}$ be the full-state mixture path (2), and define*

$$c_t := -\frac{\dot{\alpha}_t}{1-\alpha_t}, \qquad 0 < \alpha_t < 1.$$

Then for every $y \neq x$,

$$Q_t(y \mid x) = c_t\, q_{0|t}(X_0 = y \mid X_t = x) = c_t\, \mu_t(x)_y. \tag{4}$$

Proof. Fix $x_0 \in S$ and use the coupling (3). If $x \neq x_0$, then conditioning on $X_0 = x_0$ and $X_t = x$ forces $\tau \leq t$ and $\Xi = x$. Moving from t to $t-h$, the chain switches from x back to x_0 exactly when $\tau \in (t-h, t]$, so

$$\mathbb{P}(X_{t-h} = x_0 \mid X_0 = x_0, X_t = x) = \frac{\alpha_{t-h} - \alpha_t}{1-\alpha_t} = hc_t + o(h).$$

No other off-diagonal jump is possible, and if $x = x_0$ there is no off-diagonal jump at all. So the endpoint-conditioned generator is

$$Q_t^{x_0}(y \mid x) = c_t\, \mathbf{1}\{y = x_0\}\, \mathbf{1}\{x \neq x_0\}.$$

Averaging against the posterior gives

$$Q_t(y \mid x) = \sum_{x_0 \in S} Q_t^{x_0}(y \mid x)\, q_{0|t}(x_0 \mid x) = c_t\, q_{0|t}(X_0 = y \mid X_t = x)$$

for $y \neq x$. The identity with $\mu_t(x)$ follows from Proposition 1.1. ■

Equation (4) is the infinitesimal version of the exact two-stage bridge step above. It says that one-dimensional discrete diffusion is already fully distributional. There is no distinction between predicting a label distribution and predicting the full posterior over endpoint states because those are the same vector when $S = \mathcal{V}$. The same statement remains mathematically true for $S = \mathcal{V}^d$, but now the classifier would need $|\mathcal{V}|^d$ outputs and the local bridge could point toward arbitrary full sequences. That is exact and usually intractable.

Exercise 1.5 (Posterior vector or label?)**.** Stay in the one-token bridge $S = \mathcal{V}$, fix a current token x, and write

$$\mu_t(x) = \mathbb{E}[e_{X_0} \mid X_t = x].$$

1. Use (1) to show that $\mu_t(x)$ is a probability vector on $\mathcal{V}$.

2. Using (4), write the off-diagonal reverse rates $Q_t(v \mid x)$ for $v \neq x$ directly in terms of $\mu_t(x)$.
3. Explain why keeping only the argmax token $\arg\max_v \mu_t(x)_v$ is not enough to reconstruct the reverse generator.

Workspace

Exercise 1.6 (A tiny vocabulary posterior and reverse-rate computation)**.** Let $\mathcal{V} = \{A, B, C\}$, let

$$p_0(A) = \frac{1}{2}, \qquad p_0(B) = \frac{1}{3}, \qquad p_0(C) = \frac{1}{6},$$

let the reference law be uniform $r(A) = r(B) = r(C) = 1/3$, and suppose $\alpha_t = 1/4$. If we observe $X_t = B$, then:

1. Compute $K_{t|0}(B \mid A)$, $K_{t|0}(B \mid B)$, and $K_{t|0}(B \mid C)$.
2. Use Bayes' rule to compute the posterior vector
$$q_{0|t}(\cdot \mid B).$$
3. If $c_t = 2$, compute the off-diagonal reverse rates $Q_t(A \mid B)$ and $Q_t(C \mid B)$, and then determine the diagonal entry $Q_t(B \mid B)$.

Workspace

1.2 The factorized bridge on sequences

Now take sequences $x_0 \in \mathcal{V}^d$. The previous subsection already tells us what the exact sequence model would be. We would simply keep the same finite-state construction and enlarge the state space itself to $S = \mathcal{V}^d$. Then the posterior $q_{0|t}(\cdot \mid x)$ would be a distribution over complete endpoint sequences, not over individual coordinates. Nothing about posterior averaging changes in that exact model; the only change is that the simplex now has dimension $|\mathcal{V}|^d - 1$, which is conceptually clean and usually computationally hopeless.

Practical discrete diffusion therefore makes its tractability move at the level of the bridge, not at the level of posterior averaging. Write $K^{\text{full}}_{t|0}$, p^{full}_t, and $q^{\text{full}}_{0|t}$ for the exact full-state bridge, its marginal path, and its posterior. We then replace that bridge by a coordinatewise bridge $K^{\text{fac}}_{t|0}$, with its own marginal path p^{fac}_t and posterior $q^{\text{fac}}_{0|t}$. This is the modeling approximation. The exact and factorized sequence models share the same endpoints p_0 and p_1, but for $t \in (0, 1)$ they induce different intermediate marginals and different posteriors because they are genuinely different bridges. What survives unchanged is the organizing principle from the diffusion-models unit: once a bridge has been fixed, its reverse generator is obtained by averaging endpoint-conditioned bridge steps against the posterior of that same bridge.

Campbell's finite-state CTMC picture remains the reference description (Campbell et al., 2022, §3.1 and §4.2). In the factorized bridge studied here, the first-order moves switch one token at a time. Using the same differentiable nonincreasing schedule $\alpha : [0, 1] \to [0, 1]$, write

$$p_1(x) = \prod_{j=1}^{d} r_j(x_j)$$

for the product reference law. The factorized mixture path is

$$K^{\text{fac}}_{t|0}(x \mid x_0) = \prod_{j=1}^{d} \Big[\alpha_t \, \delta_{x_{0,j}}(x_j) + (1 - \alpha_t) \, r_j(x_j) \Big]. \tag{5}$$

Each coordinate independently keeps its $t = 0$ endpoint token with probability α_t or is redrawn from the

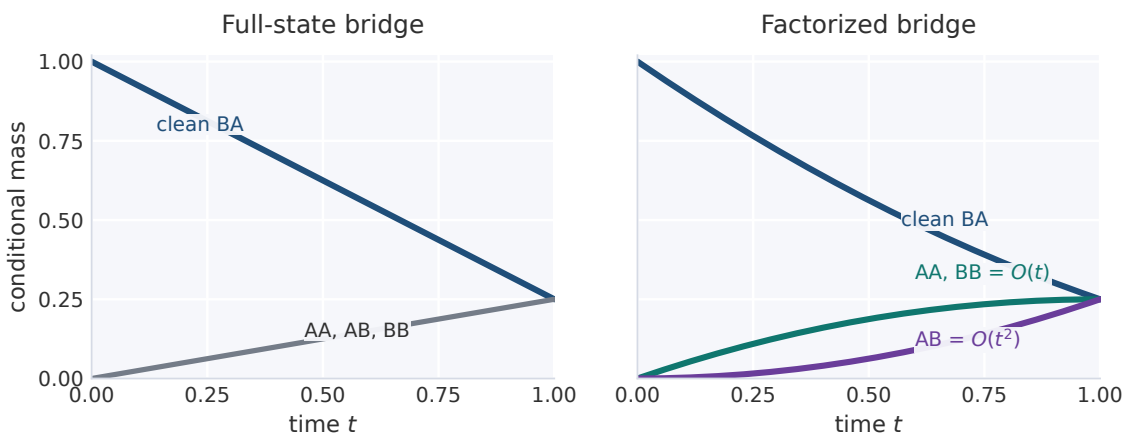

Figure 2: Condition on the current bridge state BA in a two-token binary example. Under the exact full-state bridge, the alternative endpoint sequences AA, AB, and BB all enter at first order because one bridge time controls the whole sequence. Under the factorized bridge, AA and BB require one backward switch, while AB requires two independent switches and therefore appears only at second order.

reference marginal r_j with probability $1 - \alpha_t$. Because $\alpha_0 = 1$ and $\alpha_1 = 0$, this bridge still satisfies

$$K^{\mathrm{fac}}_{0|0}(\cdot \mid x_0) = \delta_{x_0}, \qquad K^{\mathrm{fac}}_{1|0}(\cdot \mid x_0) = p_1,$$

so the factorized model keeps the same endpoint laws as the exact full-state bridge even though the intermediate path changes. The joint posterior $q^{\mathrm{fac}}_{0|t}(x_0 \mid x)$ still exists, but the local move set of the reverse dynamics has changed. To keep notation readable, from this point until the masked-diffusion subsection we again write $K_{t|0}$, p_t, and $q_{0|t}$ for the factorized bridge and its induced posterior.

A convenient coupling again makes the geometry explicit. For each coordinate j, draw an independent switch time τ_j with $\mathbb{P}(\tau_j > t) = \alpha_t$ and an independent reference token $\Xi_j \sim r_j$, and set

$$X_{t,j} = \begin{cases} X_{0,j}, & t < \tau_j, \\ \Xi_j, & t \geq \tau_j. \end{cases} \tag{6}$$

Then (6) realizes exactly the bridge (5). More importantly, it shows why the first-order generator is local. Over a short interval of length h, a single coordinate switches back to its $t = 0$ endpoint with probability $O(h)$, while simultaneous switches of two coordinates occur with probability $O(h^2)$.

This two-token toy calculation is the cleanest reason that coordinatewise marginals become sufficient after factorization. The same distinction becomes more geometric on the two-dimensional ring example from the diffusion-models unit once we discretize the plane to a grid. Bin the ring-mixture data law onto that grid, discretize the Gaussian reference law coordinatewise, and apply the factorized bridge (5). At the level of marginals we still see the same kind of interpolation from the data endpoint law to the reference endpoint law. The difference is in the sample-path geometry: the bridge no longer traces a Euclidean curve, and it instead moves by coordinatewise jumps between grid cells.

This is the discrete counterpart of the expectation-based picture in the continuous-state part of the diffusion-models unit. There one learns a conditional expectation and reads off a velocity field on $\mathbb{R}^2$. Here the same bridge logic produces local jump intensities instead of a spatial vector field, which is why the finite-state description naturally passes through a generator.

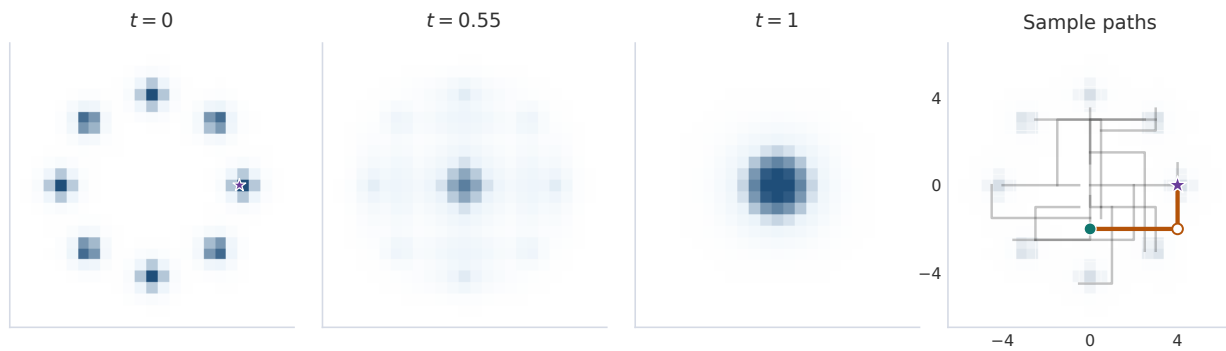

Figure 3: A discretized version of the two-dimensional ring-mixture bridge. The first three panels show the bridge marginals on the grid at $t = 0$, $t = 0.55$, and $t = 1$. The last panel shows sample paths of the factorized bridge; the highlighted path starts from the starred data cell and ends at the teal reference draw. The marginal bridge still interpolates between the same endpoint laws, but individual paths move by one-coordinate jumps.

For a sequence $x \in \mathcal{V}^d$, let $x^{j\to v}$ denote the neighbor obtained by replacing coordinate j by token v. The endpoint-conditioned generator is then as simple as possible.

Proposition 1.7 (Endpoint-conditioned backward switch rule for the factorized bridge)**.** *Let $c_t := -\dot{\alpha}_t/(1-\alpha_t)$, and assume $0 < \alpha_t < 1$. Then for every single-token neighbor $x^{j\to v}$,*

$$Q_t^{x_0}\big(x^{j\to v} \mid x\big) = c_t\, \mathbf{1}\{v = x_{0,j}\}\, \mathbf{1}\{x_j \neq x_{0,j}\}, \tag{7}$$

with diagonal chosen so that $\sum_y Q_t^{x_0}(y \mid x) = 0$.

Proof. Fix x_0 and x, and use the coordinatewise coupling (6). If $x_j = x_{0,j}$, then stepping backward does not require any change at coordinate j. If $x_j \neq x_{0,j}$, conditioning on $X_0 = x_0$ and $X_t = x$ forces $\tau_j \le t$ and $\Xi_j = x_j$, and the coordinate switches back exactly when $\tau_j \in (t-h, t]$. Therefore

$$\mathbb{P}(X_{t-h,j} = x_{0,j} \mid X_0 = x_0, X_t = x, x_j \neq x_{0,j}) = \frac{\alpha_{t-h} - \alpha_t}{1-\alpha_t} = hc_t + o(h).$$

Independence of the switch times implies that simultaneous switches of two distinct coordinates are $O(h^2)$, so they disappear at generator scale. The only off-diagonal first-order moves are the backward switches in (7). ■

To pass from the endpoint-conditioned generator to the marginal generator, we use the same posterior-averaging step as in the diffusion-models unit.

Proposition 1.8 (Generator-level marginalization on a finite state space)**.** *Suppose*

$$K_{t-h|t}(y \mid x) = \delta_x(y) + hQ_t(y \mid x) + o(h)$$

and

$$K_{t-h|t}^{x_0}(y \mid x) = \delta_x(y) + hQ_t^{x_0}(y \mid x) + o(h)$$

as $h \downarrow 0$. Then

$$Q_t(y \mid x) = \sum_{x_0 \in S} Q_t^{x_0}(y \mid x)\, q_{0|t}(x_0 \mid x).$$

Proof. Apply Proposition 1.2 with $s = t - h$, insert the first-order expansions, subtract the common zeroth-order term $\delta_x(y)$, divide by h, and let $h \downarrow 0$. ■

Theorem 1.9 (Factorized generator rates depend only on coordinate marginals)**.** *For the factorized mixture path (5), every off-diagonal single-token jump satisfies*

$$Q_t\big(x^{j\to v} \mid x\big) = c_t\, q_{0|t}\big(X_{0,j} = v \mid X_t = x\big), \qquad v \neq x_j. \tag{8}$$

Proof. Combine Proposition 1.8 and Proposition 1.7:

$$Q_t\big(x^{j\to v} \mid x\big) = \sum_{x_0 \in S} c_t\, \mathbf{1}\{x_{0,j} = v\}\, \mathbf{1}\{x_j \neq x_{0,j}\}\, q_{0|t}(x_0 \mid x).$$

If $v \neq x_j$, then the second indicator is automatically one on the event $\{x_{0,j} = v\}$, so

$$Q_t\big(x^{j\to v} \mid x\big) = c_t \sum_{x_0 \in S} \mathbf{1}\{x_{0,j} = v\}\, q_{0|t}(x_0 \mid x) = c_t\, q_{0|t}(X_{0,j} = v \mid X_t = x).$$

■

If we write

$$\mu_{t,j}(x) := \mathbb{E}[e_{X_{0,j}} \mid X_t = x] \in \mathbb{R}^{|\mathcal{V}|},$$

then Theorem 1.9 becomes

$$Q_t\big(x^{j\to v} \mid x\big) = c_t\, \mu_{t,j}(x)_v, \qquad v \neq x_j. \tag{9}$$

This is the cleanest statement of the simplification. The full posterior over endpoint sequences has not disappeared. It is still the exact posterior $q_{0|t}(x_0 \mid x)$ of the factorized bridge. What the factorized bridge changes is the local move set: once only one-token backward switches are allowed at first order, the reverse rates query that posterior only through one-coordinate marginals. So the approximation is in the bridge geometry, not in the posterior-averaging principle. This is the generator-level reason that tokenwise classification becomes the practical training target. The next theorem makes that precise: once the factorized bridge has been chosen, exact coordinate posteriors recover its exact reverse generator.

The full-state theorem above says that exact reverse kernels recover the correct law on any time grid. For the factorized bridge, the parallel statement is now infinitesimal: an ideal coordinatewise posterior model recovers the exact reverse generator of the chosen factorized bridge. Recovering the corresponding marginal path on a time grid is then a separate numerical question, handled below by a bound in terms of accumulated one-step kernel error.

Proposition 1.10 (Reverse Kolmogorov equation on a finite state space)**.** *Let S be finite, let π_t be a differentiable path of probability mass functions on S, and let Q_t be rate matrices whose entries depend*

continuously on t. Then the short-time relation

$$\pi_{t-h}(y) = \sum_{x\in S} \pi_t(x)\big[\delta_x(y) + hQ_t(y \mid x)\big] + o(h)$$

for every $y \in S$ is equivalent to the reverse Kolmogorov equation

$$-\frac{d}{dt}\pi_t(y) = \sum_{x\in S} \pi_t(x)\, Q_t(y \mid x), \qquad y \in S.$$

Given the terminal condition π_1, this linear ODE has a unique solution.

Proof. Because $\sum_x \pi_t(x)\delta_x(y) = \pi_t(y)$, the short-time relation is

$$\pi_{t-h}(y) = \pi_t(y) + h\sum_{x\in S} \pi_t(x)\, Q_t(y \mid x) + o(h).$$

Subtract $\pi_t(y)$, divide by h, and let $h \downarrow 0$ to obtain the reverse Kolmogorov equation. Conversely, if the reverse Kolmogorov equation holds, then differentiability gives

$$\pi_{t-h}(y) = \pi_t(y) - h\dot{\pi}_t(y) + o(h) = \pi_t(y) + h\sum_{x\in S} \pi_t(x)\, Q_t(y \mid x) + o(h),$$

which is the short-time relation. Uniqueness follows because this is a linear ODE on the finite-dimensional space $\mathbb{R}^{|S|}$. ■

Theorem 1.11 (Exact coordinate posteriors recover the factorized reverse generator)**.** *For the factorized mixture path (5), suppose a model returns the exact coordinate marginals*

$$\rho^\star_{t,j}(v \mid x) = q_{0|t}(X_{0,j} = v \mid X_t = x).$$

Define the learned generator by

$$\hat{Q}^\star_t\big(x^{j\to v} \mid x\big) := c_t\, \rho^\star_{t,j}(v \mid x), \qquad v \neq x_j,$$

with diagonal chosen so that $\sum_y \hat{Q}^\star_t(y \mid x) = 0$. Then $\hat{Q}^\star_t = Q_t$, where Q_t is the exact reverse generator of the factorized bridge. Consequently the factorized marginal path p_t is the unique solution of

$$-\frac{d}{dt}p_t(y) = \sum_{x\in S} p_t(x)\, \hat{Q}^\star_t(y \mid x), \qquad p_1 \text{ fixed},$$

so the exact reverse CTMC for the factorized bridge transports the factorized marginal path back to p_0.

Proof. The equality $\hat{Q}^\star_t = Q_t$ is exactly Theorem 1.9. Because Q_t is the exact reverse generator of the factorized bridge,

$$K_{t-h|t}(y \mid x) = \delta_x(y) + hQ_t(y \mid x) + o(h).$$

Combining this with Proposition 1.3 gives

$$p_{t-h}(y) = \sum_{x\in S} p_t(x)\big[\delta_x(y) + hQ_t(y \mid x)\big] + o(h).$$

By Proposition 1.10, the path p_t therefore solves the reverse Kolmogorov equation with generator $Q_t = \hat{Q}_t^\star$. Uniqueness gives the claim. ■

At this point the modeling approximation is fully visible. The full-state sequence model and the factorized sequence model are different bridges, so they have different posteriors and different intermediate marginals. But once the factorized bridge is chosen, Theorem 1.11 says that exact coordinate posteriors recover its exact reverse generator. Equivalently, within the factorized model class the marginal dynamics are completely identified by the coordinate posteriors. The remaining approximation is then only numerical: how accurately a discrete-time sampler tracks that reverse CTMC.

The exactness theorem above identifies the practical target immediately. Inside the factorized bridge, learning the reverse dynamics means learning the coordinatewise posterior marginals that determine the reverse rates. A network ρ_t^θ therefore takes a current bridge state x and time t and outputs

$$\rho_{t,j}^\theta(v \mid x) \approx q_{0|t}(X_{0,j} = v \mid X_t = x).$$

These outputs are still posterior objects: the full vector $\mu_t(x)$ in the one-token case and the coordinatewise vectors $\mu_{t,j}(x)$ in the factorized sequence case. The statistical question is whether ρ_t^θ learns the correct marginals. The numerical question, which enters only after the model is trained, is how accurately a discrete-time sampler tracks the reverse CTMC determined by those marginals. Tokenwise cross-entropy addresses the first question.

Proposition 1.12 (Cross-entropy is the proper factorized objective)**.** *Fix t, x, j, and write*

$$\pi_j(v) := q_{0|t}(X_{0,j} = v \mid X_t = x).$$

For any candidate categorical distribution $\rho(v)$,

$$\mathbb{E}_{V\sim\pi_j}[-\log \rho(V)] = H(\pi_j) + \mathrm{KL}(\pi_j \| \rho).$$

Hence the expected tokenwise negative log-likelihood is minimized uniquely at $\rho = \pi_j$.

Proof. Expand the expectation and add/subtract $\log \pi_j(v)$:

$$\mathbb{E}_{V\sim\pi_j}[-\log \rho(V)] = \sum_v \pi_j(v) \log \frac{\pi_j(v)}{\rho(v)} - \sum_v \pi_j(v) \log \pi_j(v) = \mathrm{KL}(\pi_j \| \rho) + H(\pi_j).$$

Only the KL term depends on ρ, and it is minimized uniquely when $\rho = \pi_j$. ■

The corresponding training loss is

$$\mathcal{L}_{\mathrm{disc}}(\theta) = \mathbb{E}_{X_0\sim p_0,\, X_t\sim K_{t|0}(\cdot|X_0),\, t\sim\mathrm{Unif}[0,1]} \left[\sum_{j=1}^d -\log \rho_{t,j}^\theta(X_{0,j} \mid X_t) \right], \tag{10}$$

which is exactly the discrete flow-matching objective used by Gat et al. (2024, Eq. (28) and Prop. 5). So classification is not an extra heuristic layered on top of the bridge picture. It is the proper scoring rule for the posterior statistics that the factorized generator actually needs. It therefore gives the training loop directly.

Algorithm 1 Training a factorized discrete diffusion model

Input: Data sampler for p_0, base marginals $r_1, \dots, r_d$, schedule α_t, posterior network ρ_t^θ

for training steps **do**
  Sample $x_0 \sim p_0$ and $t \sim \mathrm{Unif}[0,1]$
  for $j = 1, \dots, d$ in parallel **do**
    Sample $B_j \sim \mathrm{Bernoulli}(\alpha_t)$ and $\Xi_j \sim r_j$
    if $B_j = 1$ **then**
      Set $x_{t,j} \leftarrow x_{0,j}$
    else
      Set $x_{t,j} \leftarrow \Xi_j$
    end if
  end for
  Form $x_t = (x_{t,1}, \dots, x_{t,d})$
  Evaluate $\rho_{t,j}^\theta(\cdot \mid x_t)$ for all j
  Compute $\ell \leftarrow \sum_{j=1}^d -\log \rho_{t,j}^\theta(x_{0,j} \mid x_t)$
  Take a gradient step on ℓ
end for

Once ρ_t^θ is trained, we obtain an approximate generator by

$$\hat{Q}_t^\theta(x^{j\to v} \mid x) = c_t\, \rho_{t,j}^\theta(v \mid x), \qquad v \neq x_j.$$

An exact simulator would sample a CTMC from this generator. For intuition, the simplest approximation is a parallel Euler step, which keeps the $O(h)$ single-token events and ignores the $O(h^2)$ simultaneous jumps over a step of size h.

Algorithm 2 Euler sampling for the learned CTMC

Input: Base sampler for p_1, time grid $1 = t_K > \cdots > t_0 = 0$, schedule factor c_t, learned posterior network ρ_t^θ

Sample $X_{t_K} \sim p_1$
for $k = K, K-1, \dots, 1$ **do**
  Set $t \leftarrow t_k$, $h \leftarrow t_k - t_{k-1}$, $x \leftarrow X_{t_k}$
  Evaluate $\rho_{t,j}^\theta(\cdot \mid x)$ for all coordinates j
  for $j = 1, \dots, d$ in parallel **do**
    for each $v \neq x_j$ **do**
      $\tilde{p}_j(v \mid x_j) \leftarrow h\, c_t\, \rho_{t,j}^\theta(v \mid x)$
    end for
    $\tilde{p}_j(x_j \mid x_j) \leftarrow 1 - h \sum_{v \neq x_j} c_t\, \rho_{t,j}^\theta(v \mid x)$
    Sample $X_{t_{k-1},j} \sim \mathrm{Categorical}(\tilde{p}_j(\cdot \mid x_j))$
  end for
  Set $X_{t_{k-1}} \leftarrow (X_{t_{k-1},1}, \dots, X_{t_{k-1},d})$
end for
return X_{t_0}

Euler is only the time discretization. The structural approximation happened earlier, when we replaced the full $|\mathcal{V}|^d$-state model by the factorized local bridge. That distinction matters because classification is learning the posterior statistics needed by the chosen local generator, while Euler is only approximating

the CTMC generated by those statistics. For the per-coordinate categorical probabilities in Algorithm 2 to be valid, we choose a grid fine enough that

$$h\, c_t \le 1$$

at every step; this guarantees $\tilde{p}_j(x_j \mid x_j) \ge 0$. With that understood, the Euler sampler is one concrete family of approximate kernels $\hat{K}_k$.

This isolates the remaining numerical question. Fix a time grid and let $K_k = K_{t_{k-1}|t_k}$ denote the exact reverse kernels of the factorized bridge on that grid. Any practical sampler, including Algorithm 2, replaces K_k by approximate kernels $\hat{K}_k$. The only question is how those one-step kernel errors accumulate.

Proposition 1.13 (Accumulated one-step kernel error controls gridwise marginal error)**.** *Fix a decreasing grid $1 = t_K > \cdots > t_0 = 0$. Let $K_k = K_{t_{k-1}|t_k}$ be the exact reverse kernels of a bridge path, and let $\hat{K}_k$ be any approximate reverse kernels on the same grid. Start both chains from the same terminal law,*

$$p_K = \hat{p}_K = p_1,$$

and define

$$p_{k-1} = p_k K_k, \qquad \hat{p}_{k-1} = \hat{p}_k \hat{K}_k, \qquad k = K, \dots, 1.$$

For each step set

$$\varepsilon_k := \sup_{x \in S} \sum_{y \in S} \left| \hat{K}_k(y \mid x) - K_k(y \mid x) \right|.$$

Then for every $m = 0, \dots, K$,

$$\|\hat{p}_m - p_m\|_1 \le \sum_{k=m+1}^{K} \varepsilon_k.$$

In particular,

$$\|\hat{p}_0 - p_0\|_1 \le \sum_{k=1}^{K} \varepsilon_k.$$

Proof. If m is any signed mass function on S and K is any Markov kernel, then

$$\|mK\|_1 = \sum_{y \in S} \left| \sum_{x \in S} m(x)\, K(y \mid x) \right| \le \sum_{x \in S} |m(x)| \sum_{y \in S} K(y \mid x) = \|m\|_1.$$

So Markov kernels are ℓ_1-contractive. Now

$$\hat{p}_{k-1} - p_{k-1} = (\hat{p}_k - p_k)\hat{K}_k + p_k(\hat{K}_k - K_k),$$

and therefore

$$\|\hat{p}_{k-1} - p_{k-1}\|_1 \le \|(\hat{p}_k - p_k)\hat{K}_k\|_1 + \|p_k(\hat{K}_k - K_k)\|_1 \le \|\hat{p}_k - p_k\|_1 + \varepsilon_k.$$

Iterating backward from $\|\hat{p}_K - p_K\|_1 = 0$ gives the stated bound for every grid point. ■

Corollary 1.14 (Small-step recovery of the factorized marginal path)**.** *Consider the factorized bridge (5), and suppose the learned posterior model is ideal so that $\rho^\star_{t,j}(v \mid x) = q_{0|t}(X_{0,j} = v \mid X_t = x)$. For each grid $\Pi^{(n)} = \{1 = t^{(n)}_{K_n} > \cdots > t^{(n)}_0 = 0\}$, let $K^{(n)}_k$ be the exact reverse kernels of the factorized bridge and let*

$\hat{K}_k^{(n)}$ be any approximate kernels built from those exact coordinate marginals. If

$$\sum_{k=1}^{K_n} \sup_{x \in S} \sum_{y \in S} \left| \hat{K}_k^{(n)}(y \mid x) - K_k^{(n)}(y \mid x) \right| \longrightarrow 0 \qquad \text{as } n \to \infty,$$

then the approximate reverse chain converges at every grid point to the marginal path of the factorized bridge. In particular,

$$\|\hat{p}_0^{(n)} - p_0\|_1 \to 0.$$

Proof. By Theorem 1.11, the exact coordinate marginals recover the exact reverse generator of the factorized bridge. Hence the exact reverse kernels $K_k^{(n)}$ are the correct kernels for that bridge on the grid $\Pi^{(n)}$. Applying Proposition 1.13 to each grid gives the result. ■

This is the precise sense in which the practical factorized sampler becomes exact. It does not say that the factorized bridge reproduces the exact full-state bridge on $\mathcal{V}^d$; that modeling approximation was made earlier. It says that once the factorized bridge is chosen, exact coordinate posteriors plus vanishing one-step kernel error recover its marginal path, and therefore its data endpoint p_0, in the many-step limit.

Exercise 1.15 (Reading factorized reverse rates)**.** Stay inside the factorized bridge on the binary vocabulary $\mathcal{V} = \{A, B\}$, and let the current bridge state be $x = \texttt{BA}$. Suppose

$$q_{0|t}(X_{0,1} = A \mid X_t = x) = 0.6, \qquad q_{0|t}(X_{0,1} = B \mid X_t = x) = 0.4,$$

$$q_{0|t}(X_{0,2} = A \mid X_t = x) = 0.2, \qquad q_{0|t}(X_{0,2} = B \mid X_t = x) = 0.8,$$

and $c_t = 3$.

1. Compute the off-diagonal reverse rates from BA to AA and to BB.
2. Compute the diagonal entry $Q_t(\texttt{BA} \mid \texttt{BA})$.
3. Explain why no table of joint probabilities $q_{0|t}(X_{0,1}, X_{0,2} \mid X_t = x)$ appears in these rates.

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Exercise 1.16 (Why only one-token jumps survive at first order)**.** Use the factorized coupling (6). Fix two distinct coordinates $j \neq k$, and suppose both are currently away from their $t = 0$ endpoint values, so $x_j \neq x_{0,j}$ and $x_k \neq x_{0,k}$.

1. Show that
$$\mathbb{P}\big(X_{t-h,j} = x_{0,j},\ X_{t-h,k} = x_{0,k} \mid X_0 = x_0,\ X_t = x\big) = O(h^2).$$
2. Explain why this implies that two-token jumps disappear at generator scale.
3. Conclude that the generator-level reverse object only needs the coordinate marginals $q_{0|t}(X_{0,j} = v \mid X_t = x)$, not simultaneous two-token switch probabilities.

Workspace

1.3 Masked diffusion as a sharpened special case

Masked diffusion makes the factorized story even cleaner by collapsing the reference law to a point mass. It is not a new approximation layer; it is a particularly simple factorized bridge. Adjoin an absorbing token $[\text{MASK}] \notin \mathcal{V}$, set $r_j = \delta_{[\text{MASK}]}$ for every coordinate, and let

$$p_1 = \delta_{[\text{MASK}]^d}.$$

Each coordinate now either keeps the $t = 0$ endpoint token or is replaced by [MASK]. Equivalently, the factorized bridge becomes

$$K_{t|0}(x \mid x_0) = \prod_{j=1}^{d} \Big[\alpha_t\, \delta_{x_{0,j}}(x_j) + (1 - \alpha_t)\, \delta_{[\text{MASK}]}(x_j)\Big]. \tag{11}$$

This is the absorbing-mask version of the same factorized bridge picture.

Corollary 1.17 (Revealed coordinates are already solved)**.** *If $X_{t,j} \neq [\text{MASK}]$, then*

$$q_{0|t}(X_{0,j} = X_{t,j} \mid X_t) = 1$$

and

$$q_{0|t}(X_{0,j} = v \mid X_t) = 0 \qquad \text{for } v \neq X_{t,j}.$$

Proof. Under the masked bridge, a revealed token can only arise by keeping the $t = 0$ endpoint token. So any endpoint sequence compatible with X_t must satisfy $X_{0,j} = X_{t,j}$. ■

The second simplification is specific to the absorbing mask. Once the current state tells us which coordinates are revealed and which are masked, the schedule factor contributes the same scalar weight to every compatible endpoint sequence. After normalization it cancels.

Proposition 1.18 (The masked posterior does not depend on time)**.** *Let $x \in (\mathcal{V} \cup \{[\text{MASK}]\})^d$, and define*

$$U(x) := \{j : x_j \neq [\text{MASK}]\}, \qquad M(x) := \{j : x_j = [\text{MASK}]\}.$$

For the masked factorized bridge,

$$K_{t|0}(x \mid x_0) = \alpha_t^{|U(x)|}(1-\alpha_t)^{|M(x)|} \prod_{j \in U(x)} \mathbf{1}\{x_{0,j} = x_j\}. \tag{12}$$

Consequently,

$$q_{0|t}(x_0 \mid x) \propto p_0(x_0) \prod_{j \in U(x)} \mathbf{1}\{x_{0,j} = x_j\}, \tag{13}$$

so the posterior depends on the masked pattern and the revealed tokens, but not on t.

Proof. Each revealed coordinate contributes $\alpha_t \, \mathbf{1}\{x_{0,j} = x_j\}$, and each masked coordinate contributes $1 - \alpha_t$. Multiplying over coordinates gives (12). Bayes' rule then yields

$$q_{0|t}(x_0 \mid x) \propto p_0(x_0) \, \alpha_t^{|U(x)|}(1-\alpha_t)^{|M(x)|} \prod_{j \in U(x)} \mathbf{1}\{x_{0,j} = x_j\}.$$

The schedule factor does not depend on x_0, so it cancels during normalization. ■

Equivalently, once we observe a masked row x, the posterior is just the data law p_0 renormalized on the set of endpoint sequences that agree with the revealed coordinates $x_{U(x)}$. The only role of the forward schedule is to contribute the common scalar $\alpha_t^{|U(x)|}(1-\alpha_t)^{|M(x)|}$, and that scalar disappears when we normalize.

This is the posterior explanation for two familiar implementation choices. First, the network can ignore time in the purely masked setting because the target posterior is time-independent (Gat et al., 2024, Prop. 6 and App. E.8). Second, the loss can be restricted to masked positions because revealed coordinates already have degenerate posteriors. That is exactly why the absorbing mask collapses the generic bridge problem to ordinary prediction on the masked coordinates.

Exercise 1.19 (Why the schedule cancels in the masked case)**.** Fix a masked bridge state x, and let $U(x)$ and $M(x)$ denote its revealed and masked coordinates.

1. Starting from (12), show that every endpoint sequence compatible with the revealed coordinates receives the same factor
$$\alpha_t^{|U(x)|}(1-\alpha_t)^{|M(x)|}.$$
2. Use Bayes' rule to derive (13).
3. For a masked coordinate $j \in M(x)$, prove that for every token $v \in \mathcal{V}$,
$$q_{0|t}(X_{0,j} = v \mid X_t = x) = \mathbb{P}_{X_0 \sim p_0}\big(X_{0,j} = v \mid X_{0,U(x)} = x_{U(x)}\big).$$

4. Why can the loss be restricted to masked positions?

Workspace

1.4 Connection back to the Euclidean expectation lens

The bridge-first parallel with the Euclidean setting can be written explicitly:

$$\begin{aligned} \text{Euclidean bridge:} \quad & \text{learn } \mathbb{E}[\dot{X}_t \mid X_t = x] \text{ or } \mathbb{E}[X_0 \mid X_t = x]. \\ \text{Full finite-state bridge on } S: \quad & \text{learn } \mathbb{E}[e_{X_0} \mid X_t = x] = q_{0|t}(\cdot \mid x). \\ \text{Factorized bridge on } \mathcal{V}^d: \quad & \text{learn } \mathbb{E}[e_{X_{0,j}} \mid X_t = x] = q_{0|t}(X_{0,j} = \cdot \mid X_t = x) \text{ for each } j. \end{aligned}$$

The Euclidean model cannot represent the full posterior by a finite-dimensional simplex vector, so it learns the conditional expectations under that posterior that are infinitesimally sufficient for the marginal dynamics. The one-token discrete model can represent the full posterior exactly, because a one-hot expectation on a finite set is already a probability vector. The factorized sequence model makes the parallel tractability move: it keeps the posterior-averaging principle but chooses a local jump geometry whose infinitesimal rates depend only on coordinatewise posterior marginals. Within that factorized model, those marginals still recover the correct reverse dynamics and therefore the correct endpoint law p_0. That is the precise sense in which practical discrete diffusion reduces to classification.

So the clean conclusion is not that discrete diffusion secretly stops being distributional. It is the opposite. The exact finite-state story is fully distributional from the start, and tokenwise classification appears only after we decide to approximate the full sequence model by a factorized local bridge.

Exercise 1.20 (Classify the bridge)**.** For each scenario below, decide whether the tokenwise-classification reduction developed above is exact at the generator level, exact only in the masked/time-independent special case, or only approximate. In each case, identify the first-order reverse move set and the first model change you would try if that reduction fails.

1. A one-token categorical bridge with a known schedule.
2. A factorized sequence bridge with independent switch times.

3. A masked bridge where the network is trained only on masked positions.
4. A coupled bridge in which two-token moves occur at first order.

Workspace

1.5 Other discrete variants

The discussion above focused on one model class: a factorized bridge, its reverse CTMC, and the discrete flow-matching objective that learns the posterior marginals needed by that generator. That is the cleanest setup for developing the distributional picture without fragmentation.

Other named discrete diffusion models can mostly be understood as changing one of three ingredients while keeping the same bridge-first logic. The uniform multinomial kernel of Hoogeboom et al. (2021, §4) is the basic one-token categorical instance of this story. Absorbing-mask models are the sharpened special case from Section 1.3; in discrete time this is the absorbing-state construction in Austin et al. (2021, §3.1), while in the present flow-matching setup the posterior simplification is exactly the one stated in (Gat et al., 2024, Prop. 6 and App. E.8). Richer D3PM-style constructions play the same role by changing the one-token forward corruption kernel in the discrete-time model (Austin et al., 2021, §3.1–3.3); in the bridge-first view used here, that changes the local geometry while leaving the posterior-averaging principle intact.

So the real conceptual distinction is not between many separate algorithms. It is between the exact full-state bridge, the tractable factorized bridge, and the numerical schemes used to approximate reverse-time sampling once that bridge has been chosen. Carried back to the main notes, the lesson is the same as in the Euclidean diffusion story: choose a bridge, identify the conditional object that is sufficient for reverse-time transport, and then ask which tractable approximation preserves that sufficiency. The discrete setting is simply the case where, for some bridges, that conditional object is a literal posterior vector rather than a lower-dimensional summary.

References

Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. Structured denoising diffusion models in discrete state-spaces. *Advances in Neural Information Processing Systems*,

34, 2021.

Andrew Campbell, Joe Benton, Valentin De Bortoli, Thomas Rainforth, George Deligiannidis, and Arnaud Doucet. A continuous time framework for discrete denoising models. *Advances in Neural Information Processing Systems*, 35:28266–28279, 2022.

Itai Gat, Tal Remez, Neta Shaul, Felix Kreuk, Ricky T. Q. Chen, Gabriel Synnaeve, Yossi Adi, and Yaron Lipman. Discrete flow matching. *Advances in Neural Information Processing Systems*, 37:133345–133385, 2024.

Emiel Hoogeboom, Didrik Nielsen, Priyank Jaini, Patrick Forré, and Max Welling. Argmax flows and multinomial diffusion: Learning categorical distributions. *Advances in Neural Information Processing Systems*, 34, 2021.